DAVID SURA

Mechanical Engineering for Process and Equipment Development
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EDUCATION

University of Illinois Chicago Expected May 2028
B.S. in Mechanical Engineering | GPA: 4.0/4.0 | Dean's List

EXPERIENCE

Digital Twin & Spatial Data Intern Aug 2026 - Present
Reconstruct, Inc.

- Built and hardware-tested AI-assisted Android capture workflows integrating 360-degree camera controls, wireless transfers, and AWS S3 multipart uploads; validated camera-to-phone transfers up to 2.33 GB.
- Build regression tests and controlled device checks for connection recovery, file matching, and upload completion; use diagnostic logs and PostHog dashboards and session recordings to investigate failures.
- Set up Codex and Claude Code workflows with isolated Git worktrees, status files, append-only handoff logs, resource checks, and test gates for parallel development and cross-agent review.
- Implemented upload timing instrumentation and connection reuse to compare concurrent-transfer performance and isolate bottlenecks.
- Align floor-plan layouts with 3D site models using in-house software to connect 2D maps with captured facility geometry.

Undergraduate Researcher | Rheology and Flow Processes Aug 2026 - Present
University of Illinois Chicago

- Built AI-assisted Python rheometer-data workflows with 38 numerical and boundary-condition checks; quantified repeatability and transient response and delivered a flow-history analysis independently reproduced by the faculty supervisor.
- Compared four pump configurations for non-Newtonian flow research using pressure-flow capability, sample capacity, cleaning needs, and equipment-placement constraints; coordinated technical inquiries with two manufacturers.
- Synthesized 20 full-text studies into an evidence matrix and translated findings into measurement priorities, equipment requirements, and a staged flow-validation proposal.

ENGINEERING PROJECTS

Autonomous Payload Drone Fall 2025 - Present
Mechanical Lead | Autonomous Robotics and Systems, UIC

- Lead mechanical design and hardware integration; reduced measured drone mass from 15 lb to 9 lb (40%) through roughly 10 CAD/build iterations of the frame, motor mounts, and electronics packaging.
- Use SolidWorks finite-element analysis (FEA) to evaluate arm and mount deformation; compare mass, thrust, and power tradeoffs to guide propulsion and battery selection.
- Integrate six motors/ESCs, flight controls, GPS, telemetry, and onboard computing with the team; support 3D printing, laser cutting, and CNC job-shop fabrication for prototype hardware.

DasuraGo | Electric Freight Planning Simulator 2026 - Present
Independent Engineering Project

- Directed and validated AI-assisted development of a Python simulator linking vehicle forces, energy use, charging, and cargo constraints to compare electric-freight operating scenarios.

TECHNICAL SKILLS

Mechanical: SolidWorks CAD/FEA, engineering drawings, 3D printing, laser cutting, hardware integration
Analysis and test: Python, NumPy, Excel, MATLAB/Simulink, regression testing, fault isolation, data visualization
AI and software: Git worktrees, Codex, Claude Code, AWS S3, PostHog, Android development workflows